

# One $C^*$ -algebra and its representation-theoretic properties

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## Scope

The chosen homogeneity family defines one fixed unital simple infinite-dimensional algebra with a unique nonzero irreducible representation class. It also has a faithful representation on a separable trace space but no nonzero separable irreducible representation.

7 direct Cards + 54 reused prerequisites = 61 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

## Cards in this route

- Key result: A separably represented  $C^*$ -algebra with no separable irreducible representation
- Key result: A simple  $C^*$ -algebra with a unique irreducible representation class

[Read this route with prerequisites](#)

## Detailed route conditions

Specialize the supplied family to the once-chosen homogeneity family. The algebra remains  $T = C^*(\pi(A), \{L_i\})$ , with inclusion  $\rho_0$ . The structural theorem gives its faithful irreducible inclusion and the unique nonzero irreducible representation class, including possibly nonunital inputs.

The same  $T$  acts faithfully on the separable CAR trace space by  $\tilde{\rho}(t) = E^{-1}\rho_\phi(t)E$ . It has no nonzero irreducible representation on a separable Hilbert space. The proof first establishes unit preservation and composes the two classification unitaries in the direction  $U_\sigma U_\rho^{-1}$  before applying the finite-dimensionality theorem. This does not rule out its irreducible inclusion on the nonseparable atomic space.

## Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

## Level 0 (6 Cards)

Level 0

### The selected GNS representation of a pure-state class

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations

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Level 0

### Exact projection links for one approximately inner automorphism

A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family.

`MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: Choosing the CAR shell family from proved homogeneity

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Level 0

### The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · Approximately inner homogeneity of pure CAR states · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · The completed CAR algebra is infinite-dimensional · A two-sided inner intertwining sequence for two states

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Level 0

### A fixed representative family of CAR shell data

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · Shell matching fixes the trace of transported flags · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation

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| <div>Level 0</div> <div><b>The closed operator algebra generated by a representation and extra operators</b></div> <div>Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra.</div> <div>MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget</div> <div> <div>Immediate Card prerequisites: None in this selected scope</div> <div>Used by in this scope: An irreducible operator algebra constructed from projection shells · Reconstructing a represented unitary from its shells and residual corner</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                       |
| <div>Level 0</div> <div><b>The normalized trace on a finite CAR stage</b></div> <div>The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions.</div> <div>MathlibAnnex.CStarAlgebra.CAR.stageTrace</div> <div> <div>Immediate Card prerequisites: None in this selected scope</div> <div>Used by in this scope: Constructing the normalized trace on the completed CAR algebra</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                                                                                                                                                           |
| <div><b>Level 1 (9 Cards)</b></div> <div>Level 1</div> <div><b>A two-sided inner intertwining sequence for two states</b></div> <div>An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed.</div> <div>MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence</div> <div> <div>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</div> <div>Used by in this scope: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div> |
| <div>Level 1</div> <div><b>Constructing the normalized trace on the completed CAR algebra</b></div> <div>Carries compatible bounded matrix traces through the algebraic limit, normed union and completion.</div> <div>MathlibAnnex.CStarAlgebra.CAR.trace</div> <div> <div>Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages</div> <div>Used by in this scope: Shell matching fixes the trace of transported flags · Extending the CAR trace to the fixed shell target</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                              |

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| <p>Level 1</p> <p><b>The product-vector state on the completed CAR algebra</b></p> <p>Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.rootState</code></p>                                                                                   |
| <p>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</p> <p>Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state</p>                                                                                                                                          |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                                                                   |
| <p>Level 1</p> <p><b>The completed CAR algebra is infinite-dimensional</b></p> <p>Rules out finite dimension by retaining matrix subalgebras of unbounded dimension.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional</code></p>                                                                                                 |
| <p>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</p> <p>Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · An irreducible CAR representation has no nonzero compact image</p>                                                                               |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                                                                   |
| <p>Level 1</p> <p><b>An irreducible operator algebra constructed from projection shells</b></p> <p>Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers.</p> <p><code>MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel</code></p> |
| <p>Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators</p> <p>Used by in this scope: The atomic-shell construction on selected pure-GNS fibers</p>                                                                                                                                       |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                                                                   |
| <p>Level 1</p> <p><b>Distinct GNS classes have no unitary intertwiner</b></p> <p>The class index makes the selected pure-GNS family pairwise unitarily inequivalent.</p> <p><code>MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation</code></p>                                                                    |
| <p>Immediate Card prerequisites: The selected GNS representation of a pure-state class</p> <p>Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers · Assembling selected GNS cyclic subspaces into an isometric source representation</p>                   |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                                                                   |

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| <div>Level 1</div> <div><b>The decreasing root projections in the CAR completion</b></div> <div>The distinguished stage corners give a nested sequence of projections supporting the root state.</div> <div>MathlibAnnex.CStarAlgebra.CAR.rootFlag</div> <div>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</div> <div>Used by in this scope: Root compression becomes scalar in norm · Shell matching fixes the trace of transported flags · Reconstructing a represented unitary from its shells and residual corner</div> <div>Open Declaration Card   Read exact source</div> |
| <div>Level 1</div> <div><b>Approximately inner homogeneity of pure CAR states</b></div> <div>The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms.</div> <div>MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity</div> <div>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</div> <div>Used by in this scope: Two-sided inner sequences imply pure-state homogeneity</div> <div>Open Declaration Card   Read exact source</div>                                                                           |
| <div>Level 1</div> <div><b>A finite-row average centralizes its matrix stage</b></div> <div>Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage.</div> <div>MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage</div> <div>Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages</div> <div>Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path</div> <div>Open Declaration Card   Read exact source</div>                     |

## Level 2 (5 Cards)

Level 2

### Root compression becomes scalar in norm

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · Simplicity of the completed CAR algebra · A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

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Level 2

### Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range

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Level 2

### The atomic-shell construction on selected pure-GNS fibers

Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction.

`MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel`

Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells

Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra

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| <div>Level 2</div> <div> <b>Shell matching fixes the trace of transported flags</b> </div> <div> Recover exact trace values from the initial and final supports of shell links. </div> <div> <code>MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag</code> </div> <div> <div> <div>Immediate Card prerequisites:</div> <div>The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · A fixed representative family of CAR shell data</div> </div> <div> <div>Used by in this scope:</div> <div>Transported flags vanish on vectors generated by a trace vector</div> </div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <div>Level 2</div> <div> <b>Two-sided inner sequences imply pure-state homogeneity</b> </div> <div> The forward and inverse limits close the homogeneity argument once an intertwining sequence is supplied for every pure-state pair. </div> <div> <code>MathlibAnnex.CStarAlgebra.CAR.homogeneity_of_innerIntertwiningSequences</code> </div> <div> <div> <div>Immediate Card prerequisites:</div> <div>Approximately inner homogeneity of pure CAR states · A two-sided inner intertwining sequence for two states</div> </div> <div> <div>Used by in this scope:</div> <div>Pure-state homogeneity of the completed CAR algebra</div> </div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <div>Level 3 (5 Cards)</div> <div> <div>Level 3</div> <div> <b>The atomic direct sum of the selected CAR representations</b> </div> <div> All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. </div> <div> <code>MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation</code> </div> <div> <div> <div>Immediate Card prerequisites:</div> <div>The selected GNS representation of a pure-state class · Purity of the completed root state</div> </div> <div> <div>Used by in this scope:</div> <div>Realizing a shell family by a faithful irreducible operator algebra · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation</div> </div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div> <div> <div>Level 3</div> <div> <b>Simplicity of the completed CAR algebra</b> </div> <div> Every nonzero closed two-sided ideal of the CAR algebra contains the identity. </div> <div> <code>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit</code> </div> <div> <div> <div>Immediate Card prerequisites:</div> <div>Root compression becomes scalar in norm</div> </div> <div> <div>Used by in this scope:</div> <div>An irreducible CAR representation has no nonzero compact image</div> </div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div> </div> </div> |

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| <p>Level 3</p> <p><b>The matching GNS fiber retains exactly its cyclic line</b></p> <p>Identifies the residual projection of a transported CAR flag in its matching pure-state representation.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne</code></p>             |
| <p>Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data</p> <p>Used by in this scope: The atomic common range is one embedded GNS line</p> |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                             |
| <p>Level 3</p> <p><b>Transported flags vanish on vectors generated by a trace vector</b></p> <p>Turns decay of projection traces into norm convergence on each source-orbit vector.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero</code></p>                               |
| <p>Immediate Card prerequisites: Shell matching fixes the trace of transported flags</p> <p>Used by in this scope: The target represented on the CAR trace GNS space</p>                                                                                                                                   |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                             |
| <p>Level 3</p> <p><b>An inequivalent GNS fiber has no residual common range</b></p> <p>Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero</code></p>                 |
| <p>Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data</p> <p>Used by in this scope: The atomic common range is one embedded GNS line</p>      |
| <p><b>Open Declaration Card</b>   <b>Read exact source</b></p>                                                                                                                                                                                                                                             |



## Level 4 (4 Cards)

Level 4

### The atomic common range is one embedded GNS line

Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum.

`MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span`

Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion · Realizing a shell family by a faithful irreducible operator algebra

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Level 4

### Assembling selected GNS cyclic subspaces into an isometric source representation

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

`MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry`

Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations · A fixed representative family of CAR shell data

Used by in this scope: The cyclic sum fills every irreducible target representation

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Level 4

### Reconstructing a represented unitary from its shells and residual corner

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

`MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`

Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators

Used by in this scope: The target represented on the CAR trace GNS space · An irreducible target representation has a surviving fixed space

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Level 4

### An irreducible CAR representation has no nonzero compact image

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

`MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional

Used by in this scope: Lifting a corner involution to an ambient CAR unitary · Realizing a finite Gram matrix in a represented CAR corner

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## Level 5 (4 Cards)

Level 5

### Lifting a corner involution to an ambient CAR unitary

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

`MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path

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Level 5

### Realizing a shell family by a faithful irreducible operator algebra

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers

Used by in this scope: The ambient inclusion of the fixed shell-family target · The CAR source map into the fixed shell-family target · The  $C^*$ -algebra generated by the CAR representation and the shell unitaries

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Level 5

### An irreducible target representation has a surviving fixed space

Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.

`MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace`

Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: A common root vector realizes every selected state through the generators

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Level 5

### Realizing a finite Gram matrix in a represented CAR corner

The infinite-dimensional root corner contains an isometric copy of any finite family of vectors.

`MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Purifying a vector state on a finite CAR stage

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## Level 6 (5 Cards)

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| <div>Level 6</div> <div><b>A common root vector realizes every selected state through the generators</b></div> <div>Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states.</div> <div>MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily</div> <div>Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space</div> <div>Used by in this scope: The cyclic sum fills every irreducible target representation</div> <div>Open Declaration Card   Read exact source</div>                              |
| <div>Level 6</div> <div><b>Purifying a vector state on a finite CAR stage</b></div> <div>Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation.</div> <div>MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage</div> <div>Immediate Card prerequisites: Realizing a finite Gram matrix in a represented CAR corner</div> <div>Used by in this scope: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set</div> <div>Open Declaration Card   Read exact source</div>             |
| <div>Level 6</div> <div><b>Exact vector transport along an almost central unitary path</b></div> <div>Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path.</div> <div>MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt</div> <div>Immediate Card prerequisites: Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage</div> <div>Used by in this scope: Cross-representation state approximation with a protected finite set</div> <div>Open Declaration Card   Read exact source</div> |

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| <div>Level 6</div> <div><b>The CAR source map into the fixed shell-family target</b></div> <div>Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target.</div> <div>MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom</div> <div> <div>Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra</div> <div>Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Extending the CAR trace to the fixed shell target</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                             |
| <div>Level 6</div> <div><b>The ambient inclusion of the fixed shell-family target</b></div> <div>The same concrete target has a literal representation on the selected atomic Hilbert space.</div> <div>MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion</div> <div> <div>Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra</div> <div>Used by in this scope: The ambient representation of the fixed algebra · Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                                                                  |
| <div><b>Level 7 (5 Cards)</b></div> <div>Level 7</div> <div><b>Approximating another vector state along a unitary path</b></div> <div>A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector.</div> <div>MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx</div> <div> <div>Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage</div> <div>Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div> |
| <div>Level 7</div> <div><b>The shell-generated algebra is not an algebra of all compact operators</b></div> <div>Excludes an injective representation of the infinite-dimensional unital algebra onto all compact operators.</div> <div>MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget</div> <div> <div>Immediate Card prerequisites: The CAR source map into the fixed shell-family target · The completed CAR algebra is infinite-dimensional</div> <div>Used by in this scope: Structural properties of one shell-generated C*-algebra</div> </div> <div> <div>Open Declaration Card</div> <div>Read exact source</div> </div>                                                                                                   |

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| <p>Level 7</p> <p><b>Cross-representation state approximation with a protected finite set</b></p> <p>Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx</code></p>     |
| <p>Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Exact vector transport along an almost central unitary path</p> <p>Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences</p>                                                                                  |
| <p><b>Open Declaration Card    Read exact source</b></p>                                                                                                                                                                                                                                                                      |
| <p>Level 7</p> <p><b>Extending the CAR trace to the fixed shell target</b></p> <p>The source trace first becomes a state on the existing target algebra.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace</code></p>                                                                                   |
| <p>Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Constructing the normalized trace on the completed CAR algebra</p> <p>Used by in this scope: The GNS representation of the target trace extension</p>                                                                                |
| <p><b>Open Declaration Card    Read exact source</b></p>                                                                                                                                                                                                                                                                      |
| <p>Level 7</p> <p><b>The cyclic sum fills every irreducible target representation</b></p> <p>Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source.</p> <p><code>MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry</code></p> |
| <p>Immediate Card prerequisites: A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation</p> <p>Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion</p>                     |
| <p><b>Open Declaration Card    Read exact source</b></p>                                                                                                                                                                                                                                                                      |

## Level 8 (3 Cards)

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| <div>Level 8</div> <div><b>Every irreducible representation is unitarily equivalent to the inclusion</b></div> <div>Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target.</div> <div>MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent</div>                                                                                            |
| <div>Immediate Card prerequisites: The atomic common range is one embedded GNS line · The cyclic sum fills every irreducible target representation</div> <div>Used by in this scope: Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption</div>                                                                                               |
| <div>Open Declaration Card   Read exact source</div> <div>Level 8</div> <div><b>Pure CAR states admit two-sided inner intertwining sequences</b></div> <div>Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence.</div> <div>MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure</div> |
| <div>Immediate Card prerequisites: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set · A two-sided inner intertwining sequence for two states</div> <div>Used by in this scope: Pure-state homogeneity of the completed CAR algebra</div>                                                                                        |
| <div>Open Declaration Card   Read exact source</div> <div>Level 8</div> <div><b>The GNS representation of the target trace extension</b></div> <div>The chosen state on the existing target supplies its cyclic GNS representation.</div> <div>MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation</div>                                                                                                      |
| <div>Immediate Card prerequisites: Extending the CAR trace to the fixed shell target</div> <div>Used by in this scope: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation</div>                                                                                                                                                                          |
| <div>Open Declaration Card   Read exact source</div>                                                                                                                                                                                                                                                                                                                                                          |

# Level 9 (4 Cards)

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| <div>Level 9</div> <div><b>The target represented on the CAR trace GNS space</b></div> <div>A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone.</div> <div>MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation</div> <div>Immediate Card prerequisites: The GNS representation of the target trace extension · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner</div> <div>Used by in this scope: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space</div> <div><div>Open Declaration Card</div><div>Read exact source</div></div>                                     |
| <div>Level 9</div> <div><b>Unitary equivalence without an initial unitality assumption</b></div> <div>Shows why allowing nonunital input representations does not enlarge the irreducible representation class of the shell-generated algebra.</div> <div>MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_unitaryEquivalent_nonUnital</div> <div>Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion</div> <div>Used by in this scope: Structural properties of one shell-generated C*-algebra</div> <div><div>Open Declaration Card</div><div>Read exact source</div></div>                                                                                              |
| <div>Level 9</div> <div><b>Pure-state homogeneity of the completed CAR algebra</b></div> <div>The constructed inner intertwining sequences discharge the supplier hypothesis and give an approximately inner transporting automorphism.</div> <div>MathlibAnnex.CStarAlgebra.CAR.homogeneity</div> <div>Immediate Card prerequisites: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences</div> <div>Used by in this scope: Choosing the CAR shell family from proved homogeneity</div> <div><div>Open Declaration Card</div><div>Read exact source</div></div>                                                                                                                                                         |
| <div>Level 9</div> <div><b>Closed-ideal simplicity of the shell-generated algebra</b></div> <div>Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals.</div> <div>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget</div> <div>Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion · The CAR source map into the fixed shell-family target</div> <div>Used by in this scope: Structural properties of one shell-generated C*-algebra · Faithfulness of the target-state GNS representation</div> <div><div>Open Declaration Card</div><div>Read exact source</div></div> |

## Level 10 (3 Cards)

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| <div>Level 10</div> <div><b>Choosing the CAR shell family from proved homogeneity</b></div> <div>CAR homogeneity and exact shell matching supply one family whose distinguished component is fixed by an explicit identity choice.</div> <div>MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily</div> <div><div>Immediate Card prerequisites: Purity of the completed root state · Exact projection links for one approximately inner automorphism · A fixed representative family of CAR shell data · Pure-state homogeneity of the completed CAR algebra</div><div>Used by in this scope: A simple C*-algebra with a unique irreducible representation class · The C*-algebra generated by the CAR representation and the shell unitaries</div></div> <div><div>Open Declaration Card</div><div>Read exact source</div></div> |
| <div>Level 10</div> <div><b>Structural properties of one shell-generated C*-algebra</b></div> <div>Collects the proved properties of a single generated algebra, keeping the shell-family input separate from its consequences.</div> <div>MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint</div> <div><div>Immediate Card prerequisites: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption</div><div>Used by in this scope: A simple C*-algebra with a unique irreducible representation class</div></div> <div><div>Open Declaration Card</div><div>Read exact source</div></div>                                                                                                   |
| <div>Level 10</div> <div><b>Faithfulness of the target-state GNS representation</b></div> <div>The ideal dichotomy of the same target forces the constructed nonzero representation to be injective.</div> <div>MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective</div> <div><div>Immediate Card prerequisites: The GNS representation of the target trace extension · Closed-ideal simplicity of the shell-generated algebra</div><div>Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space</div></div> <div><div>Open Declaration Card</div><div>Read exact source</div></div>                                                                                                                                                                                                   |



## Level 11 (3 Cards)

Level 11

### The $C^*$ -algebra generated by the CAR representation and the shell unitaries

Defines the single algebra  $\mathcal{T}$  used in the irreducible and separable faithful representations.

`MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra`

Immediate Card prerequisites: Choosing the CAR shell family from proved homogeneity · Realizing a shell family by a faithful irreducible operator algebra

Used by in this scope: The fixed algebra on the CAR trace Hilbert space · The ambient representation of the fixed algebra

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Level 11

### A simple $C^*$ -algebra with a unique irreducible representation class

Establishes the properties that make the fixed algebra  $\mathcal{T}$  a counterexample to Naimark's problem.

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint`

Immediate Card prerequisites: Choosing the CAR shell family from proved homogeneity · Structural properties of one shell-generated  $C^*$ -algebra

Used by in this scope: No nonzero irreducible representation on a separable Hilbert space

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Level 11

### Faithfulness of the representation on the CAR trace Hilbert space

Unitary conjugation preserves injectivity of the target-state representation.

`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective`

Immediate Card prerequisites: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation

Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space

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## Level 12 (2 Cards)

Level 12

### The fixed algebra on the CAR trace Hilbert space

Defines a representation of the same algebra  $T$  on the GNS Hilbert space of the CAR trace.

`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation`

Immediate Card prerequisites: The target represented on the CAR trace GNS space · The  $C^*$ -algebra generated by the CAR representation and the shell unitaries

Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space

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Level 12

### The ambient representation of the fixed algebra

Defines the inclusion representation  $\rho_0 : T \rightarrow B(\mathcal{H})$ .

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation`

Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · The  $C^*$ -algebra generated by the CAR representation and the shell unitaries

Used by in this scope: No nonzero irreducible representation on a separable Hilbert space

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## Level 13 (2 Cards)

Level 13

### Faithfulness of the representation on the CAR trace Hilbert space

Proves that the representation  $\tilde{\rho} : T \rightarrow B(H_\tau)$  is injective.

`MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective`

Immediate Card prerequisites: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space

Used by in this scope: A separably represented  $C^*$ -algebra with no separable irreducible representation

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Level 13

### No nonzero irreducible representation on a separable Hilbert space

Rules out irreducible representations of  $T$  on separable Hilbert spaces by the finite-dimensionality theorem for a single irreducible representation class.

`MathlibAnnex.CStarAlgebra.CAR.not_isIrreducible_of_separable`

Immediate Card prerequisites: The ambient representation of the fixed algebra · A simple  $C^*$ -algebra with a unique irreducible representation class

Used by in this scope: A separably represented  $C^*$ -algebra with no separable irreducible representation

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## Level 14 (1 Card)

Level 14

### A separably represented $C^*$ -algebra with no separable irreducible representation

The same algebra  $T$  has a faithful representation on a separable Hilbert space, but no nonzero irreducible representation on any separable Hilbert space.

`MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation`

Immediate Card prerequisites: No nonzero irreducible representation on a separable Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space

Used by in this scope: None in this selected scope

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